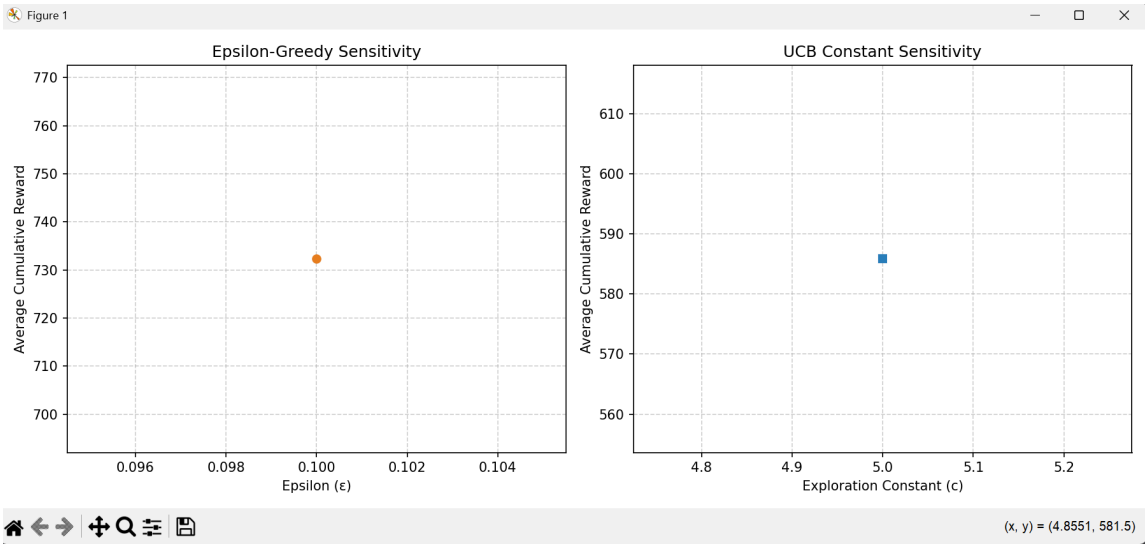
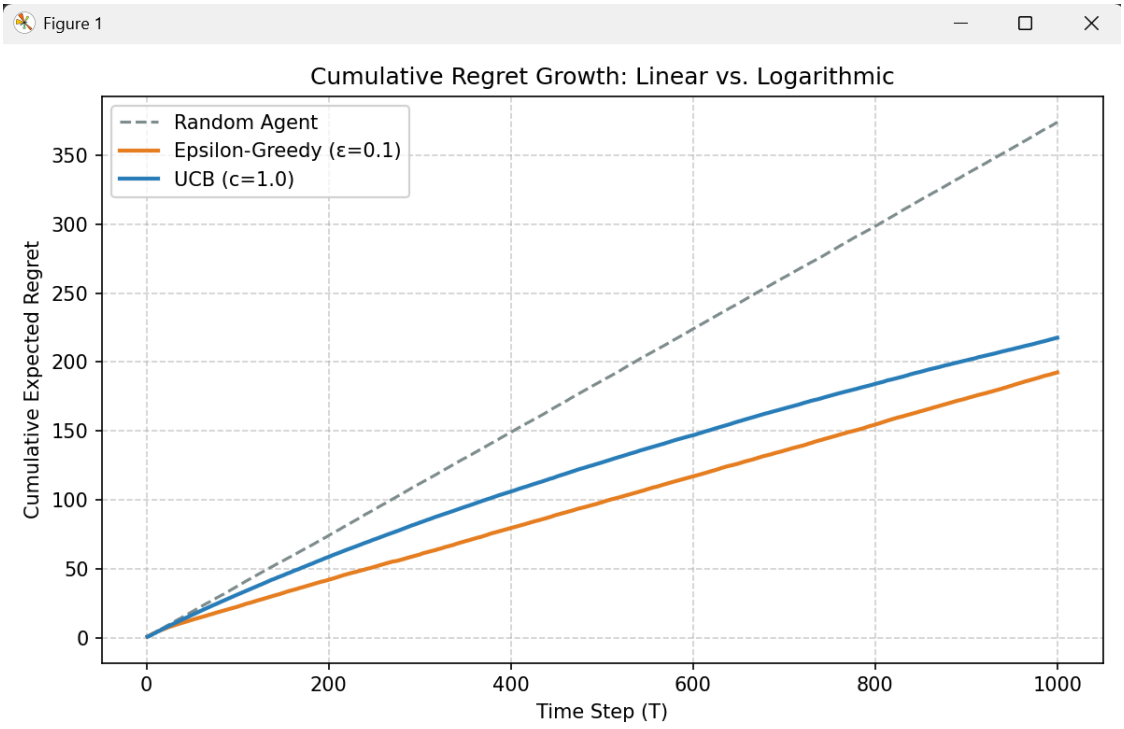
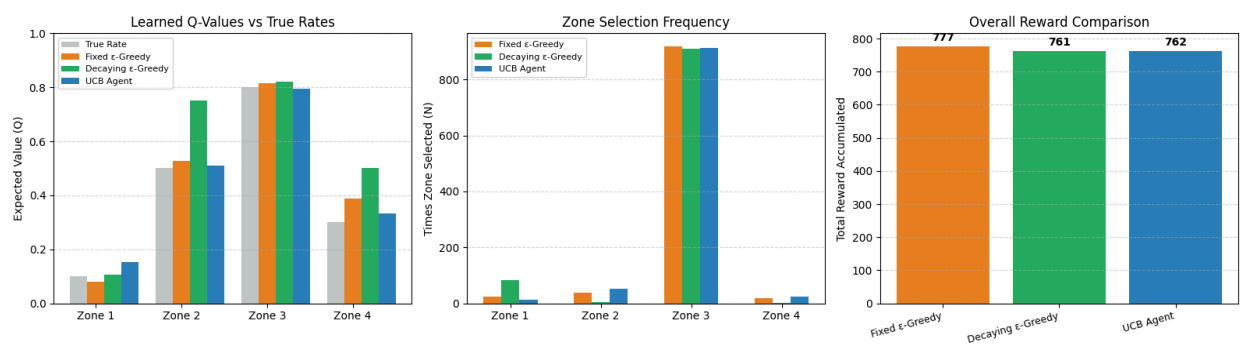
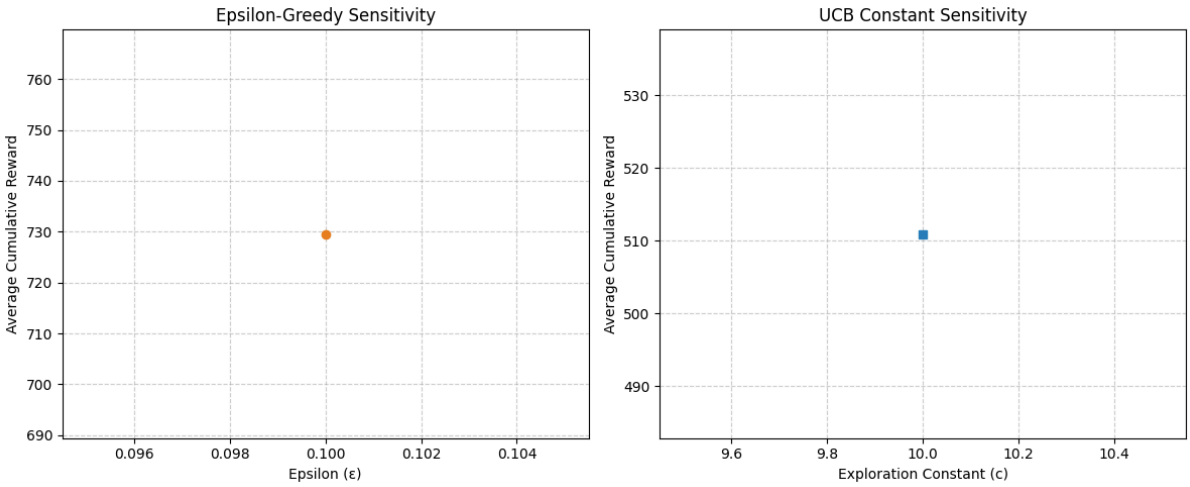
ε = 0.1; c = 5.0

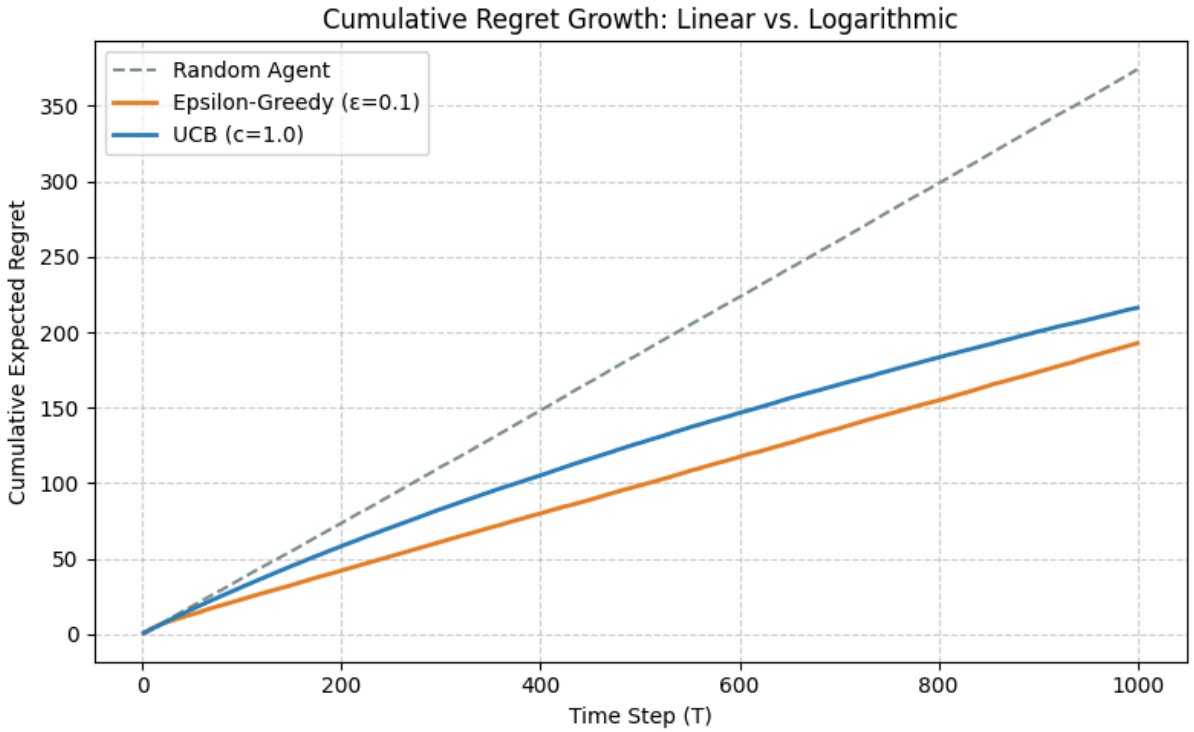

Figure 1
Epsilon-Greedy Sensitivity
Average Cumulative Reward
Epsilon (ε)
UCB Constant Sensitivity
Exploration Constant (c)
(x, y) = (4.8551, 581.5)

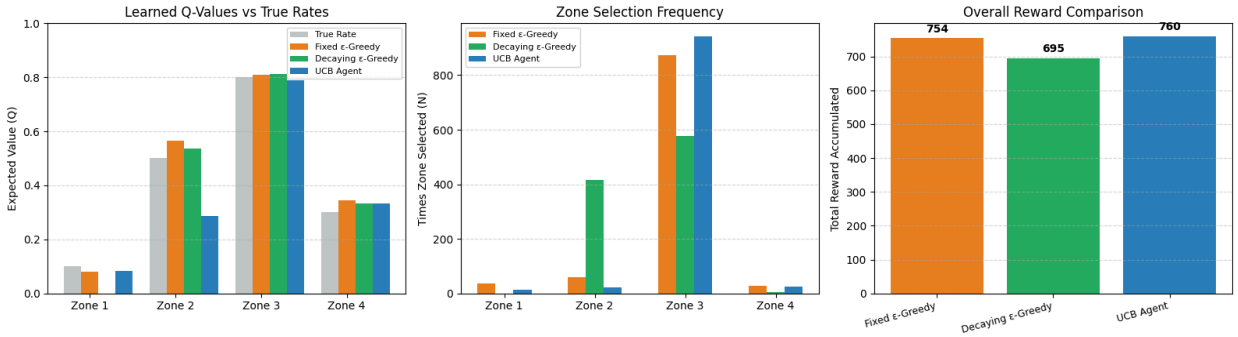

Figure 1
Cumulative Regret Growth: Linear vs. Logarithmic
Random Agent
Epsilon-Greedy (ε=0.1)
UCB (c=1.0)
Cumulative Expected Regret
Time Step (T)


Learned Q-Values vs True Rates
True Rate
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent
Expected Value (Q)
Zone 1
Zone 2
Zone 3
Zone 4
Zone Selection Frequency
Times Zone Selected (N)
Overall Reward Comparison
Total Reward Accumulated
777
761
762
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent

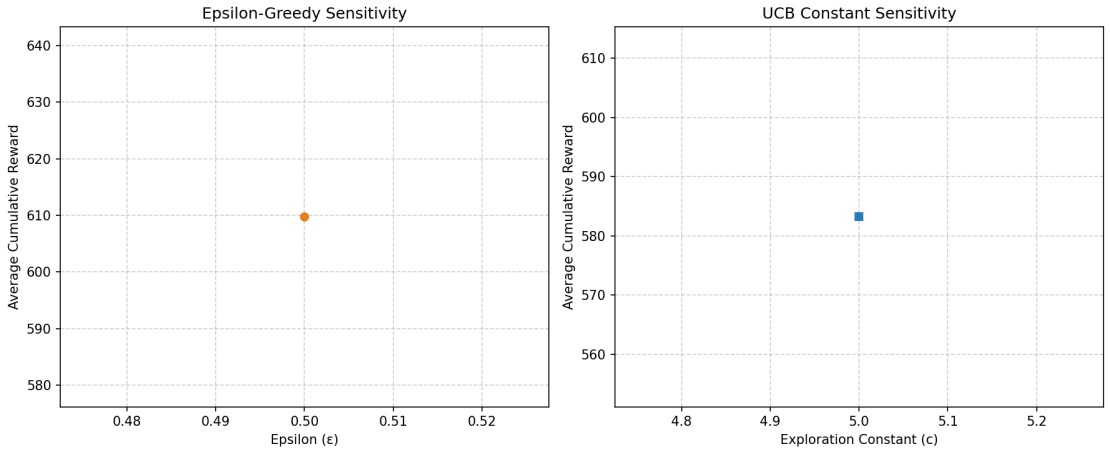
ε = 0.1; c = 10.0

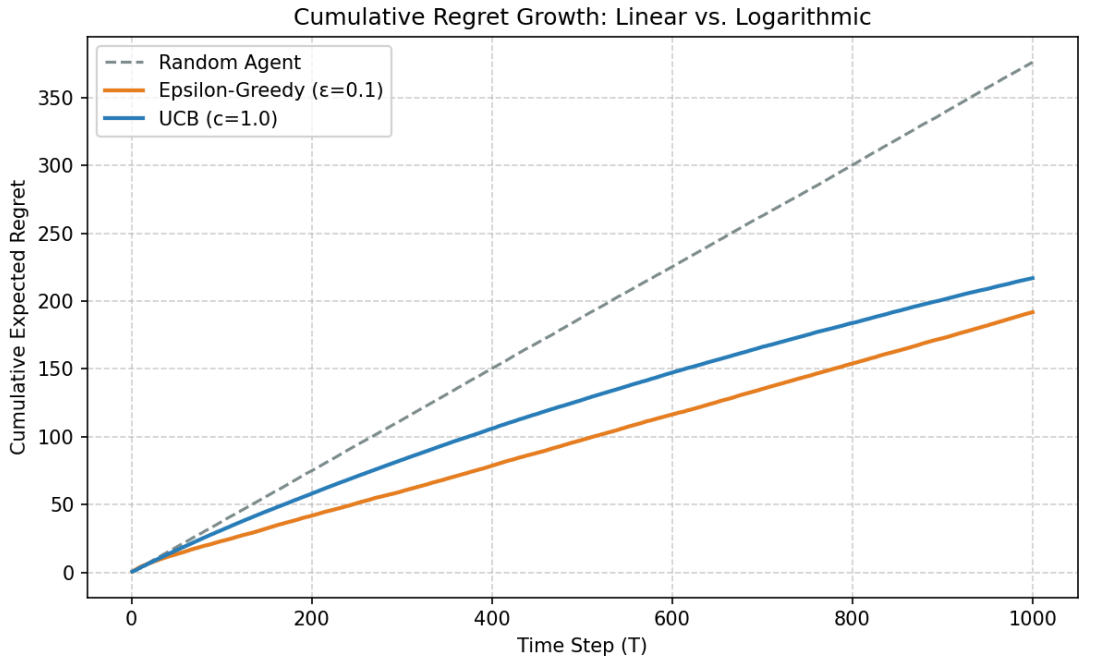

Epsilon-Greedy Sensitivity
Average Cumulative Reward
760
750
740
730
720
710
700
690
0.096
0.098
0.100
0.102
0.104
Epsilon (ε)
UCB Constant Sensitivity
Average Cumulative Reward
530
520
510
500
490
9.6
9.8
10.0
10.2
10.4
Exploration Constant (c)

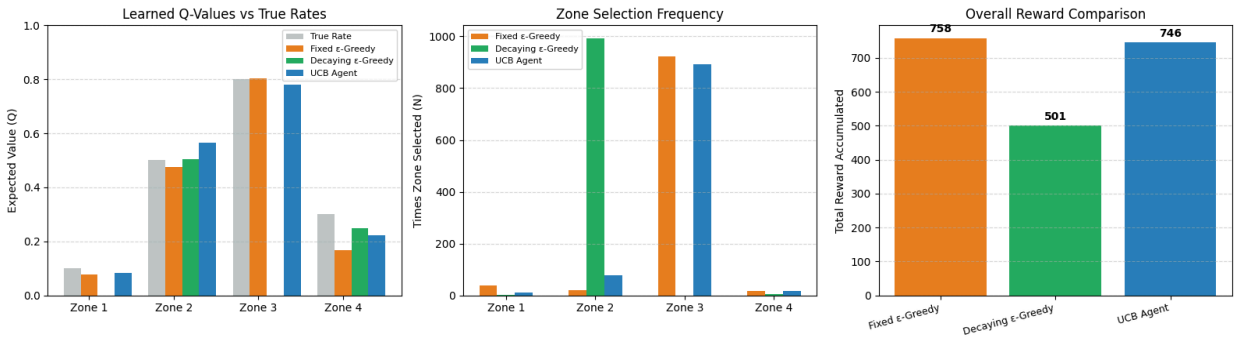

Cumulative Regret Growth: Linear vs. Logarithmic
Random Agent
Epsilon-Greedy (ε=0.1)
UCB (c=1.0)
Cumulative Expected Regret
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
Time Step (T)


Learned Q-Values vs True Rates
True Rate
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent
Expected Value (Q)
1.0
0.8
0.6
0.4
0.2
0.0
Zone 1
Zone 2
Zone 3
Zone 4
Zone Selection Frequency
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent
Times Zone Selected (N)
800
600
400
200
0
Zone 1
Zone 2
Zone 3
Zone 4
Overall Reward Comparison
Total Reward Accumulated
754
695
760
700
600
500
400
300
200
100
0
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent

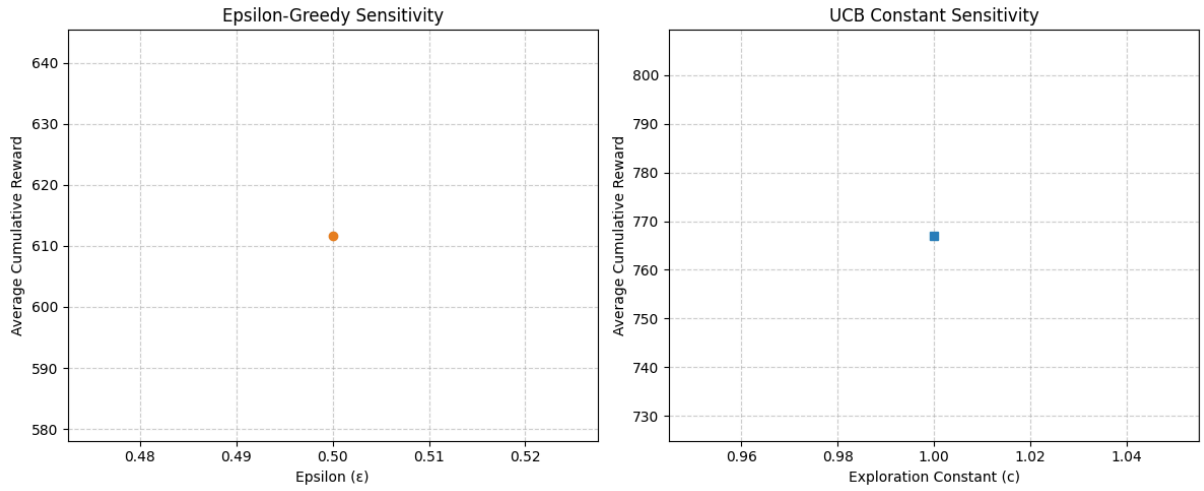
ε = 0.5; c = 5.0 (default case)

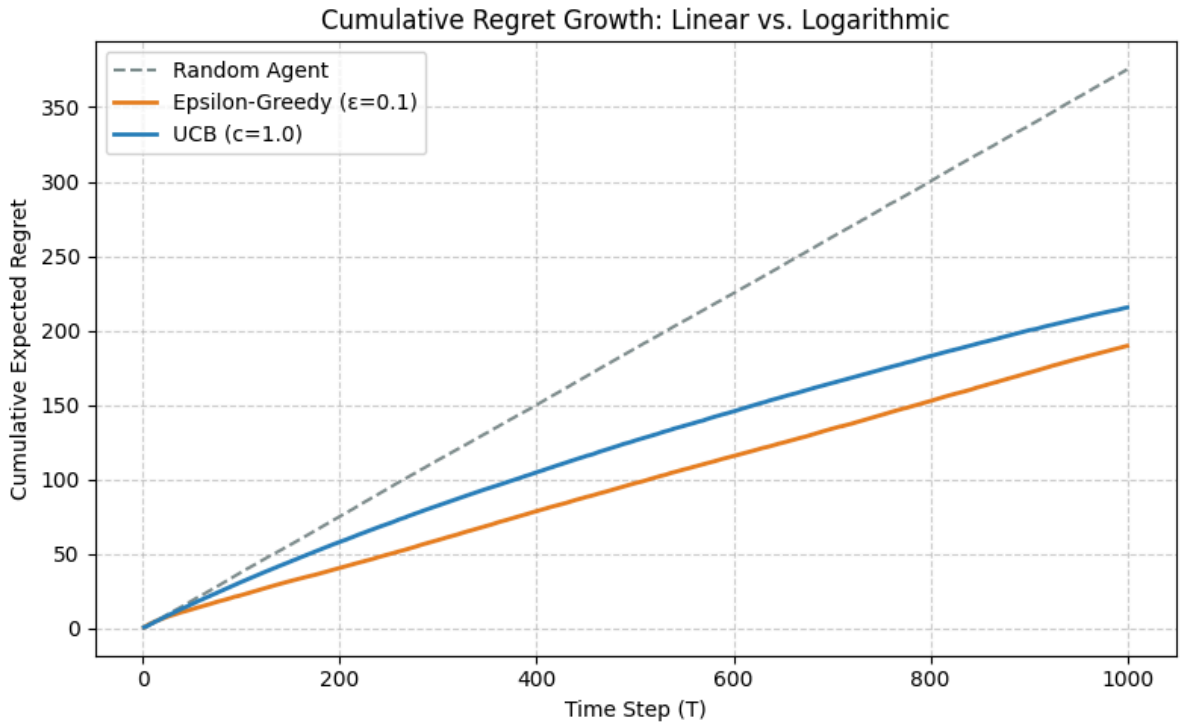


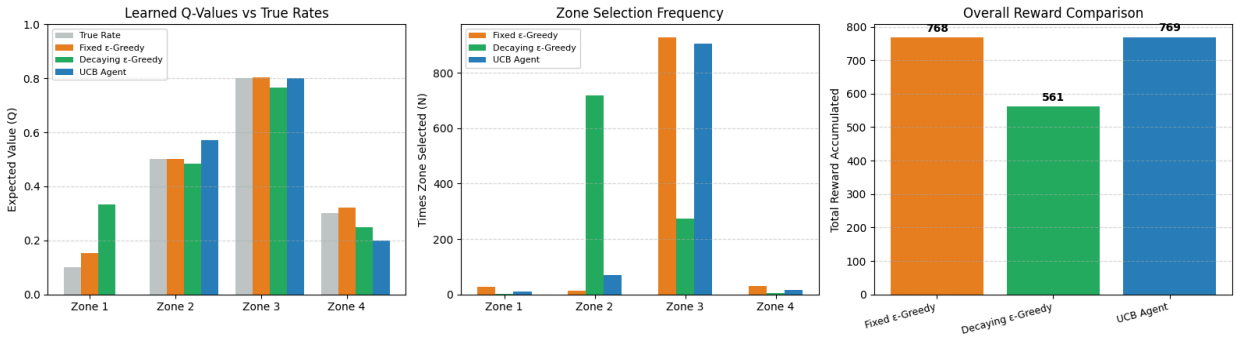




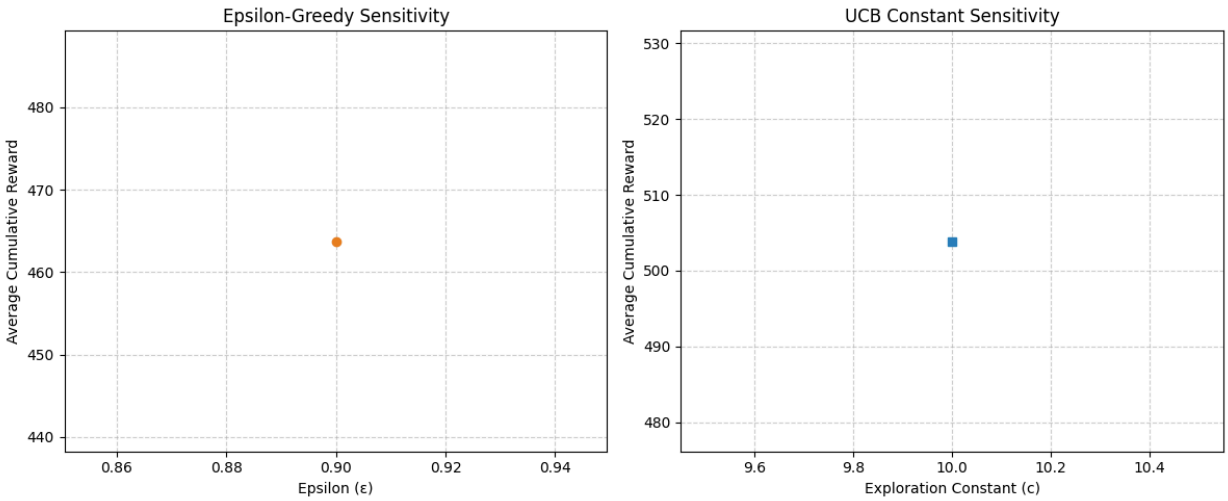
ε = 0.5; c = 1.0

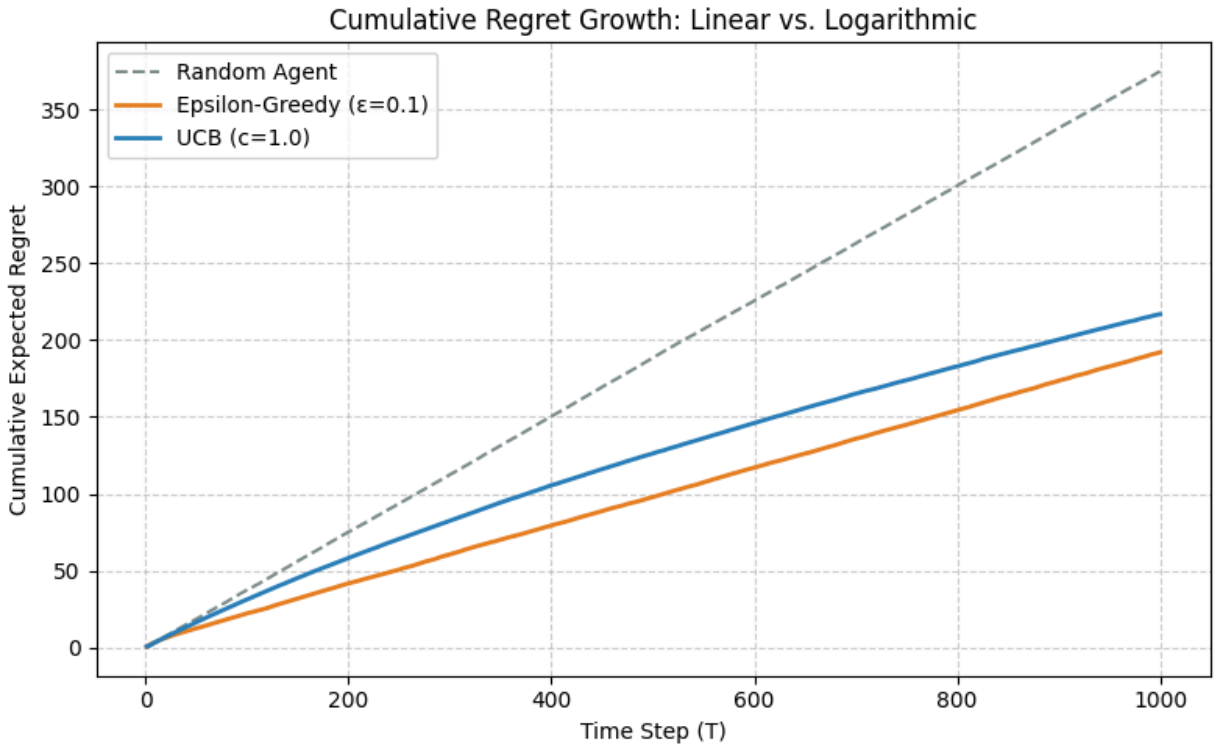

Epsilon-Greedy Sensitivity
Average Cumulative Reward
Epsilon (ε)
UCB Constant Sensitivity
Exploration Constant (c)

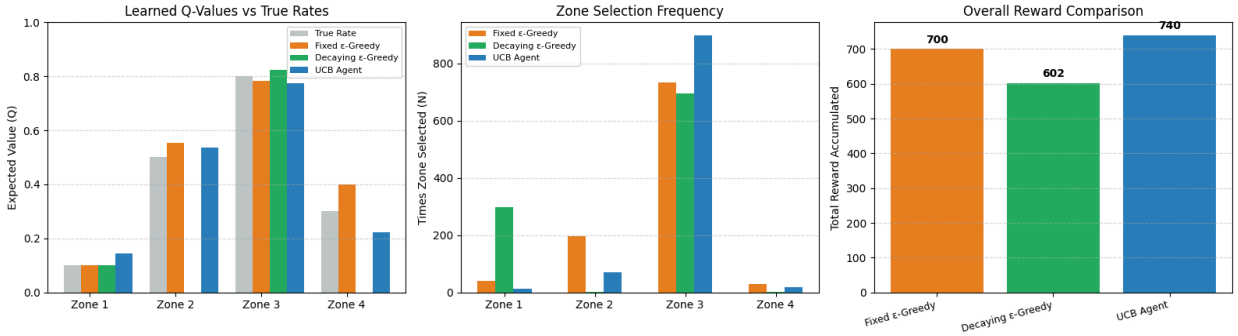

Cumulative Regret Growth: Linear vs. Logarithmic
Random Agent
Epsilon-Greedy (ε=0.1)
UCB (c=1.0)
Cumulative Expected Regret
Time Step (T)


Learned Q-Values vs True Rates
True Rate
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent
Expected Value (Q)
Zone 1
Zone 2
Zone 3
Zone 4
Zone Selection Frequency
Times Zone Selected (N)
Overall Reward Comparison
Total Reward Accumulated
768
561
769
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent

ε = 0.9; c =10.0


Epsilon-Greedy Sensitivity
Average Cumulative Reward
480
470
460
450
440
0.86
0.88
0.90
0.92
0.94
Epsilon (ε)
UCB Constant Sensitivity
Average Cumulative Reward
530
520
510
500
490
480
9.6
9.8
10.0
10.2
10.4
Exploration Constant (c)


Cumulative Regret Growth: Linear vs. Logarithmic
Random Agent
Epsilon-Greedy (ε=0.1)
UCB (c=1.0)
Cumulative Expected Regret
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
Time Step (T)


Learned Q-Values vs True Rates
True Rate
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent
Expected Value (Q)
1.0
0.8
0.6
0.4
0.2
0.0
Zone 1
Zone 2
Zone 3
Zone 4
Zone Selection Frequency
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent
Times Zone Selected (N)
800
600
400
200
0
Zone 1
Zone 2
Zone 3
Zone 4
Overall Reward Comparison
Total Reward Accumulated
700
602
740
700
600
500
400
300
200
100
0
Fixed ε-Greedy
Decaying ε-Greedy
UCB Agent

The best hyperparameter combination in terms of which leads to the highest overall reward is: ε = 0.1; c = 5.0.

Separately, the highest reward values are associated with each when ε = 0.1 (in the ε = 0.1; c = 5.0 pair) and c = 1.0 (in the ε = 0.1; c = 1.0 pair).

Thus, the combination of ε = 0.1; c = 5.0 seems to be the best overall.

Analytical Responses:

○ How does adding exponential decay to ε-greedy alter its asymptotic regret growth curve compared to fixed ε?

Adding exponential decay to the ε-greedy algorithm alters its asymptotic regret growth curve by decreasing the value of ε over time and reducing the accumulation of regret by lowering the rate of exploration (ε). Thus, it exploits more than it explores over time, which is what slows down regret accumulation since we are more closely approximating the possible optimal reward with our actual reward output.

○ In a dynamic warehouse where zone rates shift over time, why might an agent with logarithmic regret perform worse than one maintaining a small non-zero ε?

In a dynamic warehouse where zone rates shift over time, algorithms with logarithmic regret probably perform worse than the one maintaining a small non-zero ε since their regret functions decrease over time due to a decreased exploration rate as confidence improves. This becomes a problem since this does not take into account shifting rates like in a dynamic warehouse and the actions that these algorithms had come to consider optimal may have become suboptimal with these new zone rates. In this case, having a constant small non-zero ε would probably be preferred since its value does not adapt to suit its exact environment but rather it steadily explores at a specific value regardless of how many runs it does.